

A simulation model of the population dynamics of the branching coral *Acropora palmata* Effects of storm intensity and frequency

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Abstract

A stage-based model was developed to predict the effects of storm intensity and frequency on populations of the branching coral *Acropora palmata*. The data used to parameterize the model were obtained from an *A. palmata* population in the northern Florida Reef Tract that experienced three storms in 27 months.

Storm intensity is a significant factor influencing the survivorship of *A. palmata* populations. After a severe storm, *A. palmata* populations can be numerically dominated by fragments and crusts. The shift in biomass from units with high survivorship (i.e. colonies) to units with higher mortality probabilities (i.e. fragments and crusts) can affect the recovery and long-term survivorship of disturbed populations. However, the model suggests that *A. palmata* still can benefit from periodic storms. When sexual recruitment is limited or sporadic, storm fragmentation followed by fragment survivorship and growth may be the only mechanism available for *A. palmata* to propagate.

Storm frequency can also affect the survivorship of *A. palmata*. When storms occur at 15-year intervals, a slow increase in the abundance of colonies can take place after 10 consecutive storms. When storm frequency increases to 5 years, the abundance of colonies can increase five-fold after 10 storms. However, as storm frequency increases further (one storm every 2 years) a steady decline in the abundance of *A. palmata* colonies can occur.

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1. Introduction

Simulation models can provide powerful tools for the study of the dynamics of coral populations and communities. The recent concern over the rapid decline of reefs worldwide highlights the need to develop modeling tools for both scientific and management purposes (Ginsburg, 1993; Birkeland, 1997; Precht et al., 2002). Coral models such as those developed by

Hughes (1984), Done (1987), Babcock (1991), Lasker (1991), Tanner et al. (1994, 1996), Fong and Glynn (1998, 2000), and Hughes and Tanner (2000) have increased our knowledge and understanding of coral reef dynamics and have provided important forecasting tools for the prediction of the possible effects of biological and physical disturbance on reef ecosystems.

These models have shown that damage and recovery patterns of coral populations can be influenced by the characteristics of the disturbance regime (e.g. intensity, duration, frequency, timing), the characteristics of the affected coral species (e.g. growth rates,

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regeneration capabilities, reproductive mode), and the connectivity of affected populations within the local and regional landscape (Hughes and Tanner, 2000). Andres and Rodenhouse (1993) showed that resilience to physical disturbance can be closely related to coral life histories as species with fast growth rates and populations with smaller and medium-sized colonies recovered faster from storm damage. Done (1987) showed that both colony growth and recruitment rates can determine the resilience of a coral population to starfish predation locally. Finally, Johnson and Preece (1992) emphasized the importance of the spatial extent of the damage and the spatial connectivity of affected reefs.

Although models have been developed for massive, encrusting, and foliaceous corals, this study represents the first effort to model the population dynamics of a branching hard coral. *Acropora palmata* is a large, tree-like coral abundant in the shallow, wave-exposed areas of reefs throughout the Caribbean region (e.g. Goreau, 1959; Geister, 1977; Gladfelter et al., 1977; Wells, 1988). This species has experienced a significant regional decline in the past decade as a result of storms, bleaching, and epizootic diseases (Preech et al., 2002). The important role of *A. palmata* as a major reef-builder in the Caribbean region, as well as its recently documented decline, emphasize the need to understand its population dynamics.

Several unique characteristics differentiate *A. palmata* from coral species modeled in the past. Although *A. palmata* can be susceptible to physical disturbance caused by storms (Woodley et al., 1981; Lirman and Fong, 1997), it can also exhibit extraordinary regeneration and regrowth capabilities (Meesters and Bak, 1995; Lirman, 2000a). The ability of *A. palmata* to form new colonies from storm-generated fragments (Lirman, 2000b), together with the reportedly low success of sexual recruitment in this species (Bak and Engel, 1979; Hughes and Jackson, 1980, 1985; Rosesmyth, 1984), suggest a strong connection between storm disturbance and population survivorship and persistence (Lirman, 2000b).

The hypothesis that *A. palmata* may depend on storm-induced fragmentation to maintain and expand populations in the absence of significant sexual recruitment is tested here with a stage-based transition model based on data collected from the Florida Reef Tract. In addition, the model is used to evaluate the

potential impacts of different disturbance regimes (i.e. storm intensity and frequency) as well as the relative importance of sexual and asexual recruitment and partial mortality rates. The predictions made by the model are compared to field data collected from *A. palmata* populations along the Florida Reef Tract with diverse disturbance histories.

2. Methods

2.1. Model development

Unlike massive or encrusting corals where small colonies resemble adult morphologies, *A. palmata* colonies have diverse morphological stages that can be easily recognized. Here, the relative abundances of the following morphological stages characteristic of *A. palmata* are simulated.

2.1.1. Colonies

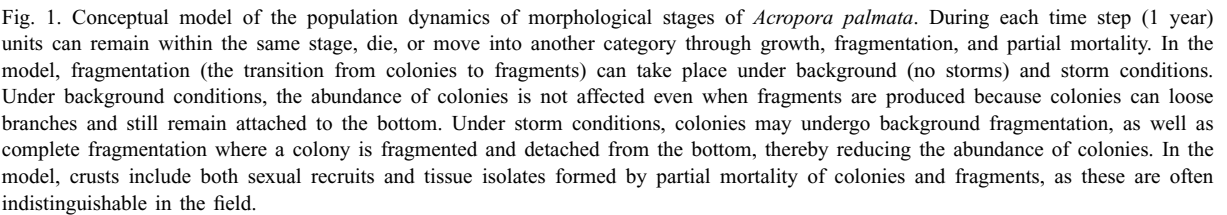
Colonies standing upright with a tree-like morphology and a single main stalk (Fig. 1). Colonies are established through sexual recruitment and the growth of fragments and crusts.

2.1.2. Fragments

Detached *A. palmata* branches with live tissue on their upper surface, but that do not exhibit vertical growth features with branching patterns (Fig. 1). Fragments can be formed by the breakage of colonies and other fragments (Lirman and Fong, 1997). *A. palmata* fragments develop upward growth features termed proto-branches usually within 1 year after fragmentation (Fong and Lirman, 1995). The transition from fragments to colonies is considered complete when proto-branches developed a branching morphology. On the average, it takes fragments 3–4 years to reach this stage after detachment (Lirman, 2000b). Tissue collections indicate that fragments can produce gametes and therefore contribute to sexual reproduction (Lirman, 2000b).

2.1.3. Crusts

Two-dimensional features found on the reef platform and upper or lower surfaces of colonies and fragments (Fig. 1). These represent die-backs of colonies and fragments formed through partial mortality, or



A stage-structured matrix model was developed to predict the relative abundance of each stage at time $t + 1$ from the population structure at time t and calculated stage transitions. The general form of this model, similar to those used by Hughes (1984), Done (1987), Lasker (1991), and Fong and Glynn (1998) is:

where x is a column vector with the population stage structure at time t and T is a matrix of stage transition probabilities calculated from field observations. During each time step ($t = 1$ year), morphological stages can remain within the same category, die, or move

A distinct feature of this model is that colonies and fragments can produce crusts (or fragments and crusts in the case of colonies) while remaining within the same stage. Branch detachment without complete colony destruction can take place during minor storms or due to boat impacts. In such cases, colonies retain some of their branches as well as their basal attachment. This is captured in the model by considering rates of “background” fragmentation ($N_{\text{colonies undergoing branch detachment}} \times N_{\text{fragments produced by each colony}}$). Accordingly, the abundance of fragments increases while the number of colonies remains the same. During storms, in addition to background or partial fragmentation, colonies can be uprooted or completely shattered. In such

Table 1

Parameters used to develop the *Acropora palmata* population model in the absence of storms

	From		
	Colonies	Fragments	Crusts
To			
Colonies	0.9	0.15	0.05
Fragments	0.1*	0.45	NT
Crusts	0.06*	0.25*	0.45
Mortality	0.1	0.4	0.5

Values represent transitions from one stage to another, except for those values with asterisk (*) in the superscript that represent the proportion of units within a stage that form fragments or crusts without undergoing total mortality. NT: no transition between the stages. Crusts include both sexual recruits and tissue isolates formed by partial mortality of colonies and fragments as these are indistinguishable in the field.

cases, the abundance of fragments increases but the abundance of colonies decreases. These transitions are captured in Tables 1 and 2. Lastly, partial mortality or fission can produce tissue isolates or

Table 2

Parameters used to develop the *Acropora palmata* population model affected by storms of (A) low, (B) medium, and (C) high intensity

	From		
	Colonies	Fragments	Crusts
To			
(A)			
Colonies	0.75	0.15	0.05
Fragments	0.15*/0.15	0.35	NT
Crusts	0.1*	0.5*	0.35
Mortality	0.1	0.5	0.6
(B)			
Colonies	0.6	0.15	0.05
Fragments	0.25*/0.25	0.25	NT
Crusts	0.15*	0.6*	0.25
Mortality	0.15	0.6	0.7
(C)			
Colonies	0.4	0.15	0.05
Fragments	0.35*/0.35	0.15	NT
Crusts	0.25*	0.8*	0.15
Mortality	0.25	0.7	0.8

Fragmentation rates (transition from colonies to fragments) include both the proportion of colonies that form fragments without undergoing total mortality (*), as well the proportion of colonies that fragment completely, becoming fragments. NT: no transition between the stages.

crusts without modifying the abundance of colonies and fragments. Crust formation rates are calculated as: $N_{\text{colonies or fragments undergoing partial mortality}} \times N_{\text{crusts formed}}$.

2.2. Model parameterization

The data used to parameterize the model were collected at Elkhorn Reef, Florida (25°21.766'N, 80°09.961'W; Fig. 2) between 1993 and 1997 following methods described by Lasker (1990, 1991) for the study of the population dynamics of branching gorgonian corals. The *A. palmata* population on this reef was impacted by Hurricane Andrew (24 August 1992; maximum sustained winds = 225 km/h; 1993), the Storm of the Century (12–13 March 1993; maximum sustained winds = 75 km/h), and Tropical Storm Gordon (14–17 November 1994; maximum sustained winds = 84 km/h) (Lirman and Fong, 1997).

A survey of 100 *A. palmata* colonies revealed that although partial loss of tissue was common, complete mortality of colonies was rare. During the 5-year observation period, only one *A. palmata* colony (50 cm in canopy diameter) experienced complete mortality due to unknown causes between 1994 and 1995.

Whenever evidence of recent breakage was discovered, the percentage of colonies showing signs of fragmentation was determined and the number of detached branches per colony estimated ($n = 50$ colonies surveyed after each storm). This information was transformed into mean fragmentation rates per colony after storms of high (Hurricane Andrew), medium (Tropical Storm Gordon), and low (Storm of the Century) intensity (Tables 1 and 2).

Marked *A. palmata* fragments ($n = 65$) were surveyed at yearly intervals to determine fragment survivorship and growth of proto-branches. Most fragments experienced some partial tissue losses but the average annual total mortality of fragments, including both removal from the reef and in situ mortality, was 45%.

The formation of crusts by fission or partial mortality, as well as crust survivorship were estimated from photographs of marked colonies ($n = 30$) and fragments ($n = 35$). Partial mortality was common on Elkhorn Reef; 6% of colonies and 25% of fragments experienced partial tissue mortality resulting in the formation of crusts. Fragments produced an average

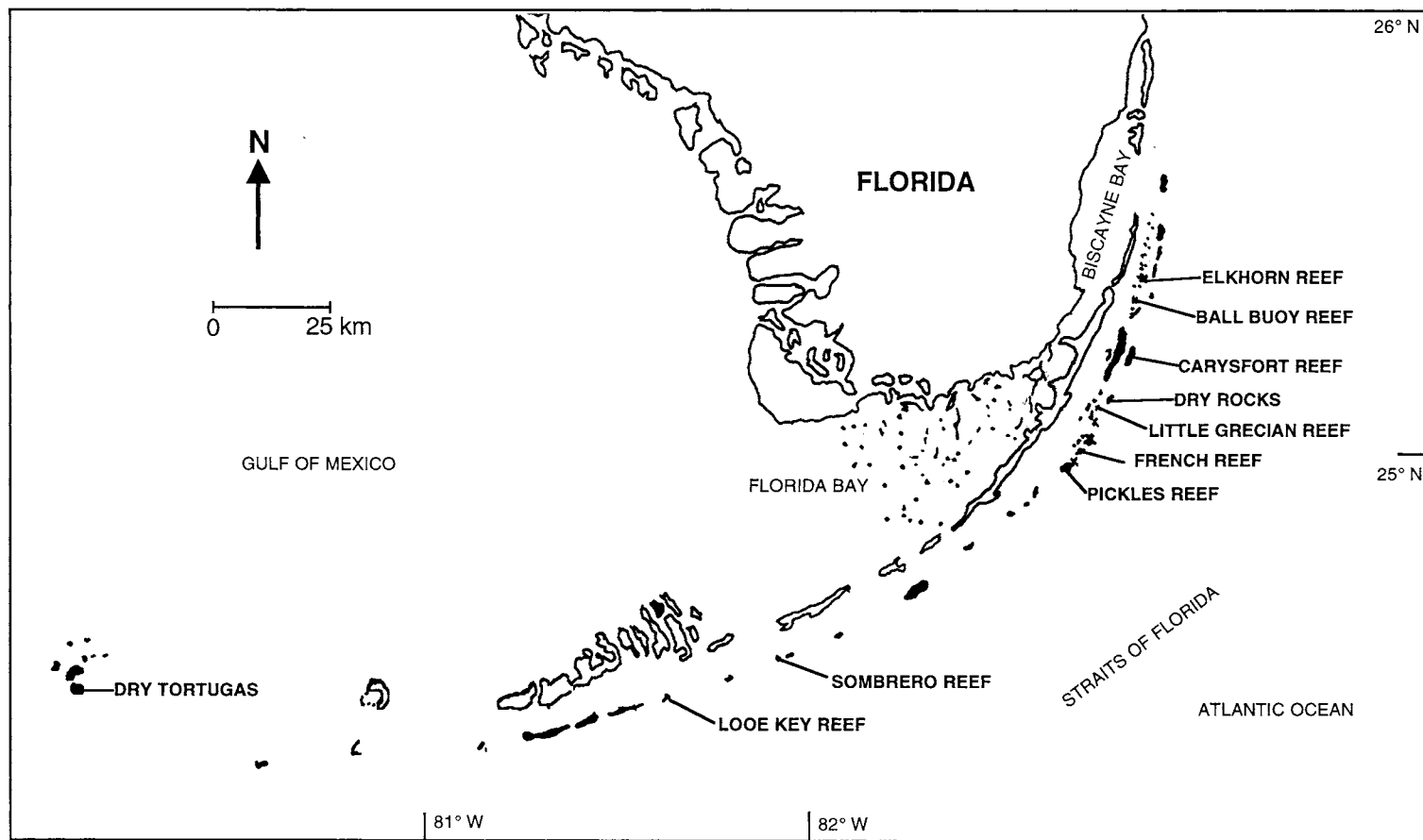


Fig. 2. Map of south Florida and the Florida Reef Tract with the location of study sites and other important reefs.

of two crusts, whereas colonies produced up to seven crusts through fission per year. Annual crust mortality in the absence of storms was 50%.

Surveys of permanent plots showed a maximum recruitment of three recruits m^{-2} per year. Since the relationship between the abundance of adult colonies and sexual recruits has not been described, recruitment in the model is assumed to be a simple linear function of the number of colonies ($N_{\text{sexual recruits per year}} = 0.1 \times N_{\text{colonies}}$).

2.3. Disturbance scenarios

2.3.1. Background conditions

To initialize each of the model runs, a 25-year simulation was performed with a starting abundance of 10 colonies (all the other categories were at zero). Fragmentation rates were low (10%), simulating background fragmentation conditions as a result of winter storms or human activities (e.g. anchor damage, boat groundings) (Table 1). This was intended to represent the conditions for an established *A. palmata* population outside the influence of hurricanes.

2.3.2. Storm effects

Storms can have significant effects on the survivorship and growth of all morphological stages of *A. palmata* (Table 2). During storms, fragmentation rates are high and colonies can be completely shattered or removed from the reef. Mortality of newly formed and pre-existing fragments by removal, burial, and scouring can be also high during storms and these negative effects were incorporated into the model as changes in transition probabilities.

The negative effects of storms on fragment survivorship can be especially high if a storm takes place within the first year after the fragments are created, as these fragments may not have had sufficient time to cement to the bottom and could be easily removed from the reef by currents and waves generated by the new storm. In the model, annual fragment mortality rates increase from 45% under background conditions to 55–75% during storms.

To evaluate the impacts of storm intensity, a single storm event of different intensity was simulated followed by a recovery period of 25 years without storms. Transition probabilities were varied according to storm intensity (Table 2). To evaluate the impacts

of storm frequency, storm events of medium intensity were simulated at different intervals (2, 5, and 15 years). The first storm was always simulated after 25 years of background conditions and a sequence of 10-storm events was simulated for each scenario.

2.3.3. Sexual recruitment and fission

Fission or partial tissue mortality rates as a result of sediment scouring, burial, and projectile movement can be high during a storm and transitions for all stages were modified according to storm intensity. Another effect of severe fragmentation is the cessation of sexual reproduction until regeneration and regrowth have been completed (Lirman, 2000b). As a function of colony abundance, sexual recruitment is reduced in the model immediately following each storm that results in colony losses. Sexual recruitment is removed completely during the storm years and on the year following the storm, and increased linearly over the following 5 years to reach the maximum rates ($N_{\text{sexual recruits per year}} = 0.1 \times N_{\text{colonies}}$).

Previous studies have highlighted the importance of asexual reproduction for the maintenance and propagation of *A. palmata* (Highsmith, 1982). Here, the relative importance of sexual and asexual reproduction were tested by altering the rates of sexual recruitment. In addition, the sensitivity of the model to changes in fission rates was evaluated.

2.4. Model performance

The *A. palmata* model was developed under the assumption that different disturbance events create changes in the relative abundances of the morphological stages that can be recognized in field surveys. For example, a clear sign of a recent hurricane would be the presence of fragmented *A. palmata* colonies and fragments deposited on the bottom. In contrast, an outbreak of coral predators or a coral disease can cause significant tissue mortality and increased rates of fission (Gladfelter, 1982; Knowlton et al., 1988) that can result ultimately in a high relative abundance of crusts.

The model was used to predict the relative abundance of each morphological stage under different disturbance scenarios. Model output was compared to data collected at several reefs of the Florida Reef Tract: Ball Buoy Reef, Key Largo Dry Rocks, Little Grecian

Reef, French Reef, and Pickles Reef (Fig. 2). On each reef, 50 m² belt transects ($n = 3\text{--}4$) were established haphazardly within *A. palmata* patches and the number of units belonging to each morphological stage were counted.

No damage to reefs in the Upper Florida Keys was reported in the 25 years prior to the surveys. Therefore, background disturbance scenarios (i.e. no major storms) were simulated for reefs of the Key Largo area (Key Largo Dry Rocks, Little Grecian Reef, French Reef, and Pickles Reef). For the two reefs from the northern Florida Reef Tract, Elkhorn Reef and Ball Buoy Reef, no significant storm damage was reported prior to 1992 when Hurricane Andrew, the Storm of the Century, and Tropical Storm Gordon impacted *A. palmata* populations on these reefs (Lirman and Fong, 1997). Accordingly, a disturbance scenario with a storm of high intensity (Hurricane Andrew), followed by a storm of low intensity (Storm of the Century) within 1 year, and a second storm of medium intensity (Tropical Storm Gordon) within 2 years was simulated for these reefs. The output from the simulations were compared to the proportion of each morphological stage measured at these reefs.

3. Results

3.1. Background conditions

After an initial decrease in the number of colonies and corresponding increases in the abundance of fragments and crusts, the abundance of each morphological type remained stable under background conditions in the absence of storms or other major disturbances (see the first 25 years in Fig. 3).

3.2. Storm intensity

Storm intensity can influence the abundance of each of the morphological stages of *A. palmata* populations (Fig. 3). The immediate effect of a single storm was to increase the number of fragments and decrease the number of colonies. The number of crusts increased only after storms of medium and high intensity. Stronger storms resulted in a larger decrease in the number of colonies, as well as a larger increase in the number of fragments and crusts.

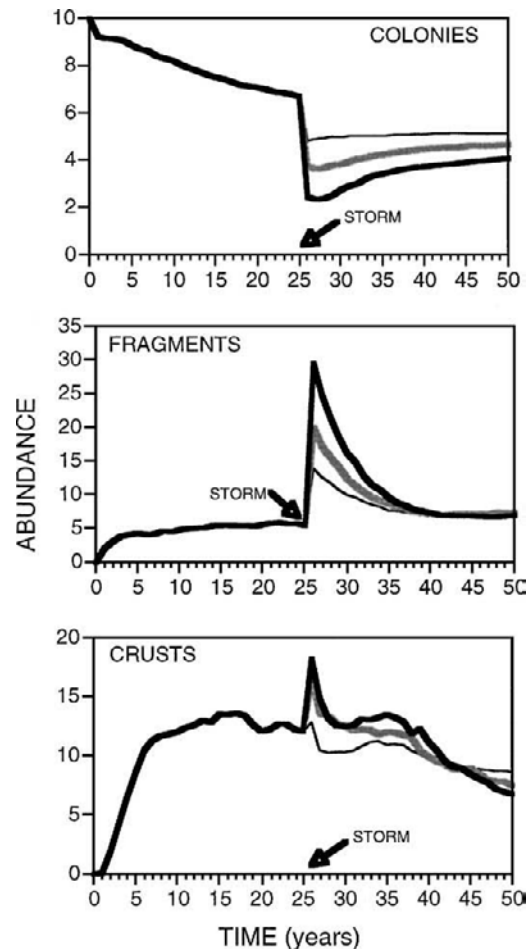


Fig. 3. Changes in the abundance of *Acropora palmata* stages in response to storms of different intensities (low intensity: thin black line; medium: grey line; high: black bold line). All the simulations were initiated with 10 colonies. Single storms were simulated after 25 years of background conditions.

The relative contribution of each stage to an *A. palmata* population can be clearly affected by the storm intensity (Fig. 4). Following a storm of high intensity, over 50% of the population can be composed of fragments, and the percentage of standing colonies can be reduced to <5%.

3.3. Storm frequency

When storms were simulated at 15-year intervals, decreases in colony abundance caused by each storm

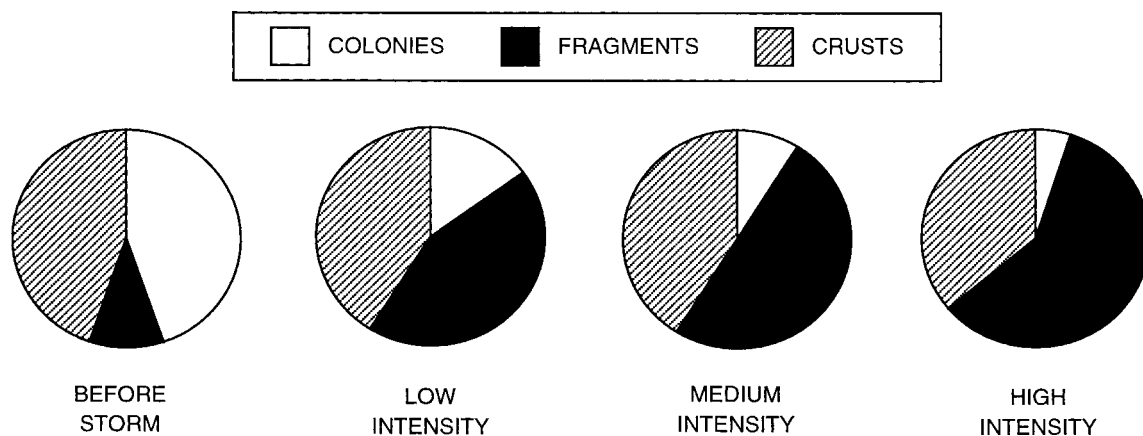


Fig. 4. Effects of storm intensity on the stage-composition of *Acropora palmata* populations prior to the storm and immediately after storms of different intensity.

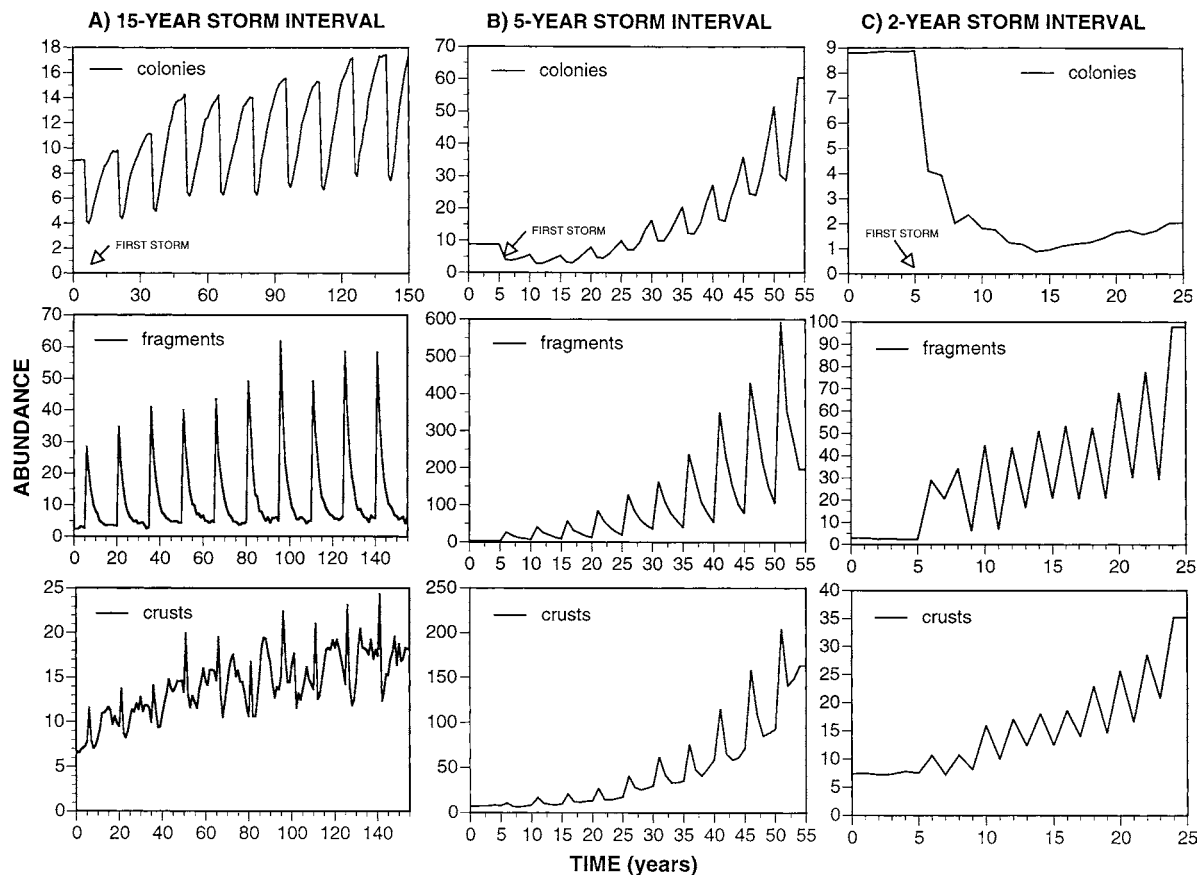


Fig. 5. Changes in the abundance of *Acropora palmata* stages in response to storms of medium intensity at (A) 15-year interval, (B) 5-year interval, and (C) 2-year interval. Background conditions were simulated for 25 years prior to the first storm. Note differing X- and Y-axis scales among the different scenarios.

were followed by recovery between storms (Fig. 5A). The abundance of colonies in the time interval prior to the new storm exceeded the maximum abundance recorded before the preceding storm event. In the absence of subsequent storms, the abundance of colonies reached the mean levels that exceeded the initial abundance prior to the first storm. A similar pattern of increase after each storm event was observed for crusts.

Fragment abundance increased after each storm followed by declines between storms. Unlike colonies and crusts that increased their abundance after each storm to levels higher than those preceding the last disturbance, fragments did not show such increasing

patterns, remaining at levels similar to those prior to the first storm.

Similar to what was observed for colonies and crusts in the 15-year storm frequency scenario, when storm frequency increases to 5 years, there was an increasing trend in the abundance of all morphological units (Fig. 5B). In all cases, abundances at the end of the 5-year recovery period between storms exceeded those recorded prior to the previous storm. However, unlike the previous scenario, colonies and fragments exhibited a sharp, increasing trend with time. When storm frequency was further increased to 2 years, fragments and crusts showed the increasing trends observed in previous storm scenarios but the abundance of colonies

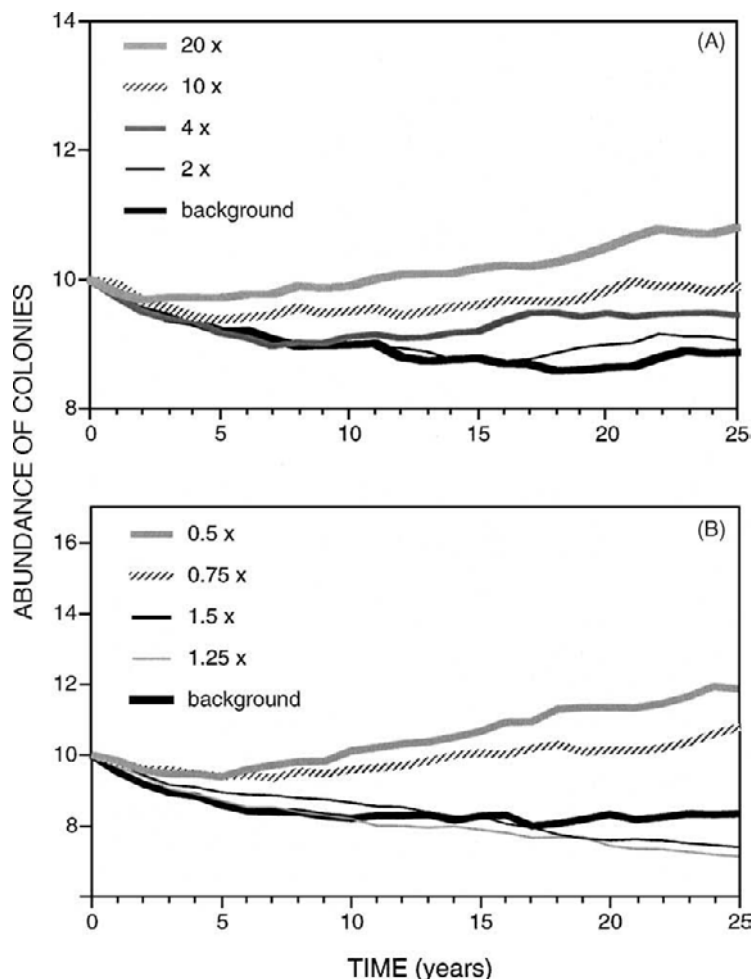


Fig. 6. Abundance of colonies with increasing (A) sexual recruitment and (B) fission or partial mortality rates. All the simulations were started with 10 colonies.

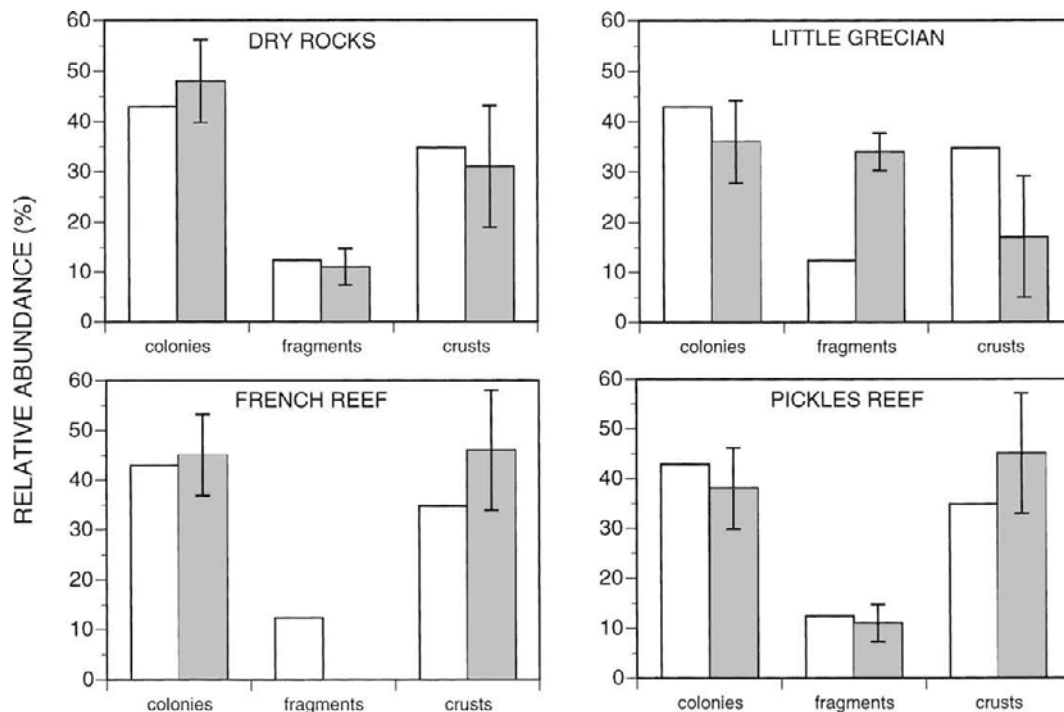


Fig. 7. Comparison between model output (white bars) and field observations (grey bars) of the mean proportion (± 1 S.E.M.) of *Acropora palmata* stages within reefs of the Key Largo area. $n = 3-4$ transects (50 m^{-2}) per reef.

experienced a sharp drop starting with the first storm, and did not recover thereafter (Fig. 5C).

3.4. Sexual recruitment and fission

In the absence of storms, the number of standing colonies declined if sexual recruitment rates were as measured in the field, and at least a 10-fold increase in recruitment rates was needed to reverse the decline in colony abundance in the absence of storms (Fig. 6A). The model was more sensitive to changes in fission rates (Fig. 6B). Increases in fission rates of 25 and 50% above those measured resulted in a steady decline in the mean abundance of colonies, whereas decreases in the observed fission rates led to increases in the mean abundance of colonies even in the absence of storms.

3.5. Model performance

Model output was within 1 S.E.M. of measured mean proportion of all stages of *A. palmata* on Key

Largo Dry Rocks and Pickles Reef (50 m^{-2} transects, $n = 3-4$ per reef; Fig. 7). On French Reef, the model output matched the percent of colonies and crusts but overestimated the proportion of fragments which were not found on this reef. In Little Grecian Reef, field observations were within 1 S.E.M. of the model output for colonies, but the model underestimated the percentage of fragments and overestimated the proportion of crusts.

Model predictions for Elkhorn Reef were within 1 S.E.M. of the field observations for all the stages when timing and magnitude of Hurricane Andrew, the Storm of the Century, and Tropical Storm Gordon were simulated (Fig. 8). This comparison provided a good verification of model performance as all model parameters were originally estimated at this reef. However, when a storm scenario similar to that used for Elkhorn Reef was used initially for Ball Buoy Reef, model predictions differed from field observations, with fragments and crusts greatly overestimated, and colonies underestimated. Field observations after Hurricane Andrew

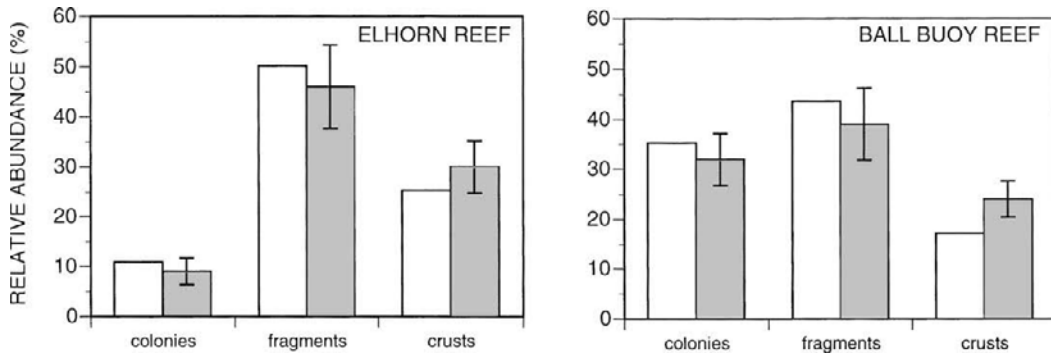


Fig. 8. Comparison between model output (white bars) and field observations (grey bars) of the mean proportion (± 1 S.E.M.) of *Acropora palmata* stages within reefs of the northern Florida Reef Tract. $n = 3\text{--}4$ transects (50 m^{-2}) per reef.

revealed that this reef, only 3 miles south of Elkhorn Reef did not exhibit similar damage levels (Porter and Meier, 1995; Lirman and Fong, 1997). *A. palmata* at Ball Buoy Reef experienced only minor fragmentation during Hurricane Andrew, when 10% of colonies showed signs of breakage, and no significant signs of damage were detected after the Storm of the Century. Considering the limited damage observed at Ball Buoy Reef, the storm scenario for this reef was modified to include just a storm of low intensity (Hurricane Andrew) followed by a storm of medium intensity (Tropical Storm Gordon) within 27 months. Using this new disturbance scenario, model predictions were within 1 S.E.M. for colonies and fragments, but underestimated the observed abundance of crusts (Fig. 8).

4. Discussion

The stage-based simulation model developed for *A. palmata* provided important insights into the population dynamics of this species with respect to physical disturbance as the composition of the coral population was shown to depend directly on disturbance history. In the absence of fragmentation caused by storms, an *A. palmata* population can reach a steady state where the proportion of units from each of the morphological stages oscillates around a constant mean. However, changes in population structure can be caused by storms, and the magnitude of the change is directly related to both storm intensity and frequency.

After a strong storm, a population can be numerically dominated by fragments and crusts, with standing colonies comprising as little as 5% of the total number of units. The significance of this in terms of the future survivorship of the affected population can not be ignored. The shift in biomass distribution from units with low mortality (i.e. colonies) towards units with higher mortality (i.e. fragments and crusts) can decrease the recovery probability of a population and severely hinder long-term survivorship. Similarly, the depletion of large colonies within a populations has been identified as a critical sign of population decline that can influence the persistence of reef-building corals (Done, 1988; Hughes and Tanner, 2000).

Coral populations can be very susceptible to subsequent disturbances after single or multiple storm events. Such an example of increased susceptibility after a major storm was documented by Knowlton et al. (1981, 1988) in a study of *Acropora cervicornis* damaged by Hurricane Allen in Jamaica. Although this population survived the storm, the remaining units, consisting mostly of fragmented colonies and tissue isolates, succumbed rapidly to delayed mortality and predation. Significant delayed mortality of *A. palmata* fragments has also been documented by Highsmith et al. (1980) and Rogers et al. (1982).

The recovery of damaged populations can be hindered by the low abundance of colonies after storms. Populations that experience significant tissue losses rely heavily on the recruitment of sexually produced larvae to repopulate the affected areas (Connell and Keough, 1985; Hughes and Tanner, 2000). Since

sexual reproduction of *A. palmata* is related to colony size and damage levels (Lirman, 2000b), a population that experiences severe fragmentation and loss of colonies will have decreased sexual reproductive potential for several years following a disturbance, effectively reducing the recovery capabilities of the affected populations (Aronson and Precht, 2001). The output from the model suggests that *A. palmata* populations that receive limited inputs of sexual recruits may depend on recurrent storm fragmentation to increase colony abundance. Under background fragmentation conditions and low sexual recruitment, *A. palmata* populations showed a decline in the number of standing colonies as a result of fission and mortality. Increases in colony abundance were not detected until sexual recruitment increased considerably.

Although *A. palmata* populations can certainly benefit from periodic storms, there appears to be a threshold frequency that, if exceeded, can cause significant decreases in colony abundance. The effects of different storm frequencies described here support the findings from previous studies that suggest a non-linear response of coral communities to increased storm frequency (Hughes, 1984, 1989; Hughes and Jackson, 1985; Woodley, 1992). Rare events can also have a significantly greater effect on the survivorship of coral populations. As storm frequency increases, only the most resilient colonies survive, and a greater proportion of smaller colonies become abundant (Witman, 1992; Andres and Rodenhouse, 1993). For *A. palmata*, larger colonies with complex branching patterns can present a larger profile to winds and currents, and become more susceptible to dislodgment and fragmentation.

The model performed well under both background and storm scenarios, predicting the relative abundance of the different morphological stages of *A. palmata* for three of the four reefs surveyed in the Upper Florida Keys. The model output differed considerably from the field observations at Little Grecian Reef, where the abundance of fragments was underestimated and the abundance of crusts highly overestimated. This shows that reefs within the same region can have different damage and recovery trajectories even when they are exposed to similar environmental conditions. The composition of the *A. palmata* population at Little Grecian Reef suggests this reef may have been affected by a storm 5–7 years prior to the surveys.

However, such a signal was not detected for nearby Key Largo Dry Rocks, French Reef, and Pickles Reef. Similarly, when the disturbance scenarios used for Elkhorn Reef were simulated for Ball Buoy Reef, model predictions did not match the field data collected at Ball Buoy Reef. Visual surveys after each storm confirmed that these reefs had experienced significantly different damage patterns from the same storms (Lirman and Fong, 1997). Highly variable storm-damage patterns were also observed for multiple sites located within a short distance of each other in Huatulco, Mexico (Lirman et al., 2001).

The model generally underestimated the relative abundance of crusts within populations. Since crusts are formed by the fission of colonies and fragments, it is possible that fission rates documented at Elkhorn Reef where data to parameterize the model were obtained, are unusually low. Previous work by Knowlton et al. (1981, 1988) and Miller (2001) showed that the formation of coral isolates can take place under intense predation pressure by corallivores such as *Hermodice carunculata* and *Coralliophila abbreviata*. Surveys at Elkhorn Reef showed that these coral predators exist in low abundances (Lirman, unpublished data), possibly resulting in the low fission rates recorded on this reef and predicted for the reefs of Key Largo.

The development of spatial, age, and stage-based transition models has shown that disturbance regimes can play a major role in the persistence and propagation of sessile organisms (Sousa, 1984). The influence of periodic disturbances on the control of population dynamics has been documented for rocky intertidal organisms (Sousa, 1985), corals (Connell and Keough, 1985), as well as terrestrial plants (e.g. Valverde and Silvertown, 1997; Batista et al., 1998; Kiviniemi, 2002). In most instances, subordinate, competitively inferior species require a certain disturbance regime for recruitment substrate to become available and superior competitors to be removed to increase in abundance (Connell, 1978; Giho and Seno, 1997; Seno and Nakajima, 1999). While competitive interactions do not appear to limit the abundance and distribution of *A. palmata*, which can dominate shallow, wave-exposed areas of reefs, this species is still dependent on periodic disturbances to maintain and expand populations through disturbance-induced asexual reproduction or fragmentation.

The stage-based simulation model described here identified the storm intensity and frequency as important factors influencing damage and recovery patterns of *A. palmata* populations. The simulations highlighted an important trade-off between increased susceptibility following a storm event due to the relative increase in the abundance of units with high mortality probabilities (i.e. fragments and crusts) and the need for this species to propagate asexually in light of its limited or sporadic sexual recruitment success. Clearly, the difference between a storm being a destructive force or an external factor that promotes asexual propagation and population expansion is often a small one, and the balance between these two aspects of the ecology of this species ultimately will influence the long-term survivorship of *A. palmata* populations already decimated by diseases and other stressors.

Lastly, simulation models commonly highlight the areas where further research is needed. In this case, although the model developed for *A. palmata* showed good predictive power, it is clear that our understanding of sexual reproduction and fission mechanisms is very limited and needs further attention to understand fully the population dynamics of this species under different disturbance scenarios.

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